Special Functions in Apple

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Gamma Function

We use the Lanczos approximation as described in Press et al. (2007), extended to work on negative numbers:

```
λz.
  { gammaln ← λz.
    {
      zz ⟜ z-1;
      c0 ← 0.999999999999997092;
      γ ← 607%128;
      coeffs ← ⟨ 57.1562356658629235
               , _59.5979603554754912
               , 14.1360979747417471
               , _0.491913816097620199
               , 0.339946499848118887e-4
               , 0.465236289270485756e-4
               , _0.983744753048795646e-4
               , 0.158088703224912494e-3
               , _0.210264441724104883e-3
```

```
                , 0.217439618115212643e-3
                , _0.164318106536763890e-3
                , 0.844182239838527433e-4
                , _0.261908384015814087e-4
                , 0.368991826595316234e-5
                >;
      ss ← (+)/([y%(zz+itof x)]`(1..14) coeffs);
      ((zz+0.5)*_.(zz+γ+0.5)-(zz+γ+0.5))+_.(√(2*π)*(c0+ss))
    };
    e:(?z≥0.5,.gammaln z,._.π-_.(sin.(π*z))-gammaln(1-z))
  }
```

Hypergeometric Functions

The hypergeometric function is given by

$$ {}_pF_q(a_1, \dots, a_p; b_1, \dots, b_q; z) = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_q)_n} \frac{z^n}{n!} $$

where $(a)_n = a(a+1)\cdots(a+n-1)$, $(a)_0 = 1$ is the rising factorial, also called the Pochhammer symbol.

We can define the rising factorial to work on real numbers like so:

```
 > :store rf [(*)/ₒ 1 (𝑓 x (x+y-1) (ℒy))]
 > :ty rf
float → float → float
```

`rf 1` is then the factorial:

```
 > rf 1'frange 1 7 7
Vec 7 [1.0, 2.0, 6.0, 24.0, 120.0, 720.0, 5040.0]
```

The hypergeometric function in Apple:

```
λa.λb.λz.
{
  rf ← [(*)/ₒ 1 (frange x (x+y-1) (ℒy))]; fact ← rf 1;
  Σ ← λN.λa. (+)/ₒ 0 (a'ɩN); Π ⇐ λa.((*)/)∴(a');
  Σ 30 (λn. {nn⟜ℝ n; (Π (λa.rf a nn) a%Π (λb. rf b nn) b)*(z^n%fact nn)})
}
```

This takes $a_1 \dots a_p$ and $b_1 \dots b_q$ as array arguments:

```
 > :yank H math/hypergeometric.🍎
 > :ty H
Vec (i + 1) float → Vec (j + 1) float → float → float
```

erf

The error function $\mathrm{erf}(z)$ satisfies the following (Weinstein, n.d.):

$$\begin{aligned}\mathrm{erf}(z) &= \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}dt \\ &= \frac{2z}{\sqrt{\pi}}{}_1F_1\left(\frac{1}{2};\frac{3}{2};-z^2\right) \\ &= \frac{2ze^{-z^2}}{\sqrt{\pi}}{}_1F_1\left(1;\frac{3}{2};z^2\right)\end{aligned}$$

The former has convergence problems (Shaw 2002). Simplifying the latter:

```
λz.
{
  ffact ← [(*)/ₒ 1 (ƒ 1 x (⌊x))];
  Σ ← λN.λa. (+)/ₒ 0 (a'ιN);
  (2%√π)*Σ 30 (λn. {nf⟜ℝn; ((_1^n)*z^(2*n+1))%(ffact nf*(2*nf+1))})
}
```

Normal Distribution CDF

The CDF for the standard normal distribution $N(0,1)$ can be calculated as $\frac{1}{2}\left(1+\mathrm{erf}\left(\frac{z}{\sqrt{2}}\right)\right)$:

```
λz.
{
  erf ← λz.
    { ffact ← [(*)/ₒ 1 (ƒ 1 x (⌊x))]
    ; Hoa ← λN.λa. {xs ← gen. z [_1*x*(z^2)] (N+1); (+)/ₒ 0 ((%)`xs (a'ιN))}
    ; (2%√π)*Hoa 30 (λn. {nf⟜ℝn; (ffact nf*(2*nf+1))})
    };
  zz ⟜ z%√2;
  0.5*(1+erf(zz))
}
```

t-Distribution CDF

For ν degrees of freedom we have (Amos 1964):

$$\frac{1}{2}+x\frac{\Gamma(\frac{1}{2}(\nu+1))}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})}{}_2F_1\left(\frac{1}{2},\frac{1}{2}(\nu+1);\frac{3}{2};-\frac{x^2}{\nu}\right)$$

for $|x| < \sqrt{\nu}$ (${}_2F_1$ converges if and only if $|z| < 1$).

Hence:

```
λx.λν.
{
  gammaln ⟜ λz. {
    zz ⟜ z-1;
    c0 ← 0.999999999999997092;
    𝛾 ← 607%128;
    coeffs ← ⟨ 57.1562356658629235
             , _59.5979603554754912
             , 14.1360979747417471
             , _0.491913816097620199
             , 0.339946499848118887e-4
             , 0.465236289270485756e-4
             , _0.983744753048795646e-4
             , 0.158088703224912494e-3
             , _0.210264441724104883e-3
             , 0.217439618115212643e-3
             , _0.164318106536763890e-3
             , 0.844182239838527433e-4
             , _0.261908384015814087e-4
             , 0.368991826595316234e-5
             ⟩;
    ss ← (+)/ ([y%(zz+itof x)]`(1..14) coeffs);
    (((zz+0.5)*_.(zz+𝛾+0.5))-(zz+𝛾+0.5))+_.((√(2*𝜋))*(c0+ss))
  };
  f21 ← λa0.λa1.λb.λz. {
    rf ← [(*)/ₒ 1 (𝑓 x (x+y-1) (⍳y))]; fact ← rf 1;
    Σ ← λlb.λub.λa. (+)/ₒ 0 (a'(lb..ub));
    Σ 0 50 (λn. {nn⟜ℝ n; rf a0 nn*(rf a1 nn%rf b nn)*(z^n%fact nn)})
  };
  0.5+x*e:(gammaln(0.5*(ν+1))-gammaln(ν*0.5))%(√(𝜋*ν))*f21 ½ ((ν+1)%2) 1.5 (_(x^2%ν))
}
```

References

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